Brain Tumor Classification from MRI Using VGG16, Vision Transformer, and a Hybrid CNN–Transformer Architecture with Grad-CAM Explainability

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ABSTRACT

Abstract-Magnetic Resonance Imaging (MRI) of the brain is a clinically vital undertaking in which false classification has dire patient prognoses. Radiologists have a lengthy process of manual interpretation and inter-observer variability in manual radiological interpretation, which explains the necessity of precise, automated classification systems. This paper will give a comparative analysis of three deep learning architectures in four-class brain tumor MRI classification namely glioma, meningioma, pituitary tumor, and no tumor. The whole Brain Tumor MRI Dataset with 7,153 images is separately trained and tested using a fine-tuned VGG16 convolutional neural network, a Vision Transformer (ViT-B/16), and a new Hybrid VGG16-ViT fusion model with an identical stratified 80/10/10 train/validation/test split. VGG16 model has a test accuracy of 98.74, ViT-B/16 has a test accuracy of 99.16 and the proposed Hybrid model has the test accuracy of 99.16, and the further benefit of converging within 15 epochs instead of 20 epochs of each standalone model. Gradient-weighted Class Activation Mapping (Grad-CAM) is incorporated in all three settings to give heatmaps that can be spatially interpreted. The hybrid design has been determined as the best design since it is a combination of the local spatial accuracy of the convolutional feature maps with the global contextual reasoning of the transformer self-attention, attaining better generalization at lower training cost. The findings indicate that CNN-Transformer feature fusion is an effective, capable, and explainable method to be used as an automated localization of brain tumors.

Index Terms Brain tumor classification, VGG16, Vision Transformer, hybrid CNNTransformer, Grad-CAM, explainable AI, MRI, transfer learning, deep learning.

1. INTRODUCTION

One of the most fatal complications in the world with neurological conditions is brain tumor and the glioma, meningoma, and pituitary adenoma differ significantly in the level of malignancy, localization, and treatment needs [1]. Timely and precise classification through MRI dictates choice of treatment pathway and has a great impact on patient survival [2]. Radiologists can manually interpret images and have a limited time, fatigue, and inter-observer variability, which makes the application of a strong automated diagnostic support system urgent [3].

Since the advent of deep learning as a viable tool, Convolutional Neural Networks (CNNs) have been the paradigm of medical image classification. Hierarchical transfer learning on architectures like VGG16, with hierarchical-based stack of 3x3 convolutional filters has proven to be highly effective in terms of feature extraction when fine-tuned on MRI-images of the brain [4]. Nevertheless, CNNs are also restricted by local receptive fields: every convolution kernel activity is based on a specified spatial area, and the model is unable to model long-range spatial relationships on the entire MRI slice [5].

The image recognition problem is restructured by Vision Transformers (ViT) [6] as a sequence modeling task, where the image is divided into fixed 16x16 patches and all patches are run through multi-head self-attention. This means of global attention facilitates the model to detect the spatially distributed features of tumor, i.e., diffuse margins, perilesional edema and bilateral asymmetry, which are not localized within the convolutional receptive fields [7]. A natural extension of the paradigms has been hybrid CNNTransformer architectures, which combine the complementary strengths of the two paradigms [8].

Clinical deployment is a prerequisite of explainability. Gradient-weighted Class Activation Mapping (Grad-CAM) by Selvaraju et al. [9] produces spatially discriminative heatmaps using a back-propagation of gradients through the final convolutional layer, which identify the locations in the MRI most important to a classification decision. The overall model configurations presented in this work allow straightforward visual validation of model reasoning, which is a condition that clinicians trust [10] with the incorporation of Grad-CAM.

The current study fills an urgent gap in the current literature: no previous research has been conducted that compares standalone CNN, standalone Transformer, and hybrid CNN-Transformer models under the same dataset, split, preprocessing, and optimizer parameters, and integrated Grad-CAM explainability in all of the aforementioned settings. The paper has the following contributions: (i) a comparative rigorous assessment of VGG16, ViT-B/16 and a Hybrid VGG16-ViT model under the same experimental conditions; (ii) a new dual-branch feature fusion system combining 256-dimensional CNN and Transformer representations into a combined 512-dimensional classification vector; (iii) Grad-CAM visualization implemented on all three model configurations; and (iv) empirical evidence of how the hybrid model has test accuracy levels that are competitive with only 25% of the training epochs and

The further structure of this paper has the following structure. Section II is a literature review. Section III presents the methodology consisting of dataset, architectures and training. Results are presented and discussed in section IV. Section V is the conclusion of the paper.

2. RELATED WORK

Pretrained CNN architectures have been used to implement transfer learning and set good benchmarks in brain tumor MRI classification. Swati et al. [4] have shown that fine-tuned VGG19 achieves 94.82% accuracy on a three-class dataset and Deepak and Ameer [11] have shown that GoogLeNet features can achieve 97.1% accuracy on a three-class dataset with SVM classification. Cheng et al. [12] used augmented features of pretrained CNNs and achieved a competitive performance on several MRI datasets. The article by Litjens et al. [5] offers an in-depth overview of deep learning-based medical imaging analysis, having made CNN transfer learning the widely used approach in all the modalities.

Transformer-based vision algorithms have proven to be competitive and even superior in the medical imaging tasks. Dosovitskiy et al. [6] found that ViT trained on large datasets is as good as CNN in performance on natural image benchmarks. Ghassemi et al. [13] used Transformer architectures on brain tumor MRI and achieved 98.6 percent accuracy with visualization of attention maps. The study by Liu et al. [14] established that Swin Transformer, which adds a hierarchical patch merging with shifted window attention, is able to perform well on medical image classification at a lower computational cost than conventional ViT.

The use of hybrid CNN to Transformer architectures has shown that a combination of local convolutional attention feature and global attention feature implies improvements over each of the two backbones. A hybrid model with an accuracy of 99.1% on brain tumor MRI was proposed by Chen et al. [8], but the study lacks any Grad-CAM integration or a controlled comparison with standalone CNNs and ViTs under the same experimental conditions. Li and Dib [15] did use explainability in a hybrid framework but did not specify convergence efficiency. Asif et al. [16] attained 99.43% using EfficientNetB1 and explainable AI methods, whereas Afshar et al. [17] surveyed the deep learning methodology of brain tumor classification and brain tumor generation. The current paper is characterized by a full three-way controlled comparison with Grad-CAM with all settings and it exhibits more rapid convergence of the hybrid model as a measurable efficiency benefit.

3.Literature Review

Brain tumor classification using Magnetic Resonance Imaging (MRI) is a critical area in neuro-oncology, directly influencing diagnosis, treatment planning, and patient prognosis [**1**]. Recent advancements in deep learning, particularly hybrid Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), alongside Explainable AI (XAI) techniques, have significantly enhanced the accuracy and interpretability of these classifications [**1][2][3]**. This survey delves into the current literature, focusing on key aspects such as hybrid CNN-Transformer models, specific architectures like ViT-B/16 and VGG16, transfer learning, MRI, Grad-CAM, explainable AI, and real-time inference.

3.1 Hybrid CNN-Transformer Architectures for Brain Tumor Classification

The heterogeneous nature of tumor morphology and the complexity of MRI data present significant challenges for accurate brain tumor classification [**4][5]**. To address these issues, hybrid models combining the strengths of CNNs and Transformers have emerged as a promising solution [**1][6][5][7][8]**. CNNs are adept at extracting local spatial

features from images due to their inherent inductive biases of locality and translation invariance [**1][5][9]**. They process images by sliding small, learnable kernels to create feature maps, identifying fundamental visual primitives like edges and textures. However, CNNs often struggle with capturing long-range dependencies and global contextual information within an image [**10][11]**.

In contrast, Vision Transformers (ViTs), initially developed for natural language processing, process images by dividing them into patches and utilizing a self-attention mechanism to capture global relationships from the very first layer [citation:27, wiki_4, wiki_8]. This enables ViTs to model long-range interactions effectively, making them suitable for tasks requiring broad contextual understanding, wiki_4, wiki_8]. The combination of CNNs and ViTs leverages the CNN's efficiency in local feature extraction with the Transformer's strength in global reasoning [citation:7, 13, wiki_4]. For instance, one study proposed a deep hybrid framework integrating CNNs, Transformers, and XAI for classifying low-grade glioma (LGG), high-grade glioma (HGG), tumors, and healthy classes using multi-contrast MRI modalities [**1]**. Another approach, BRAIN-META, combined ten hybrid CNN-ViT models with a meta-learning ensemble framework for multi-class brain tumor classification using 2D MRI images [**6]**. A hybrid deep learning framework has been proposed that synergizes CNNs for localized spatial feature extraction and ViTs for capturing long-range contextual information, followed by an attention-guided fusion mechanism [**5]**. This synergy often leads to improved diagnostic performance, particularly on complex medical images, 13, wiki_4].

3.2 Vision Transformer (ViT-B/16) and VGG16 in Brain Tumor Classification

Specific deep learning architectures like Vision Transformers (ViTs) and VGG16 have been extensively utilized in brain tumor classification. ViTs, such as ViT-B/16, excel at capturing global contextual information, which is crucial for identifying subtle patterns across an entire MRI scan that might indicate a tumor, 34, wiki_4]. Studies have shown that ViTs can achieve superior performance compared to traditional CNN-based models, with one study reporting a 92.59% accuracy in binary classification of brain tumors using MR images [**12]**. Another work proposed ViTBrain, an MRI-based brain tumor image detection model using Vision Transformer, highlighting its ability to address the limitation of CNNs in capturing long-range information [**11]**. However, ViTs generally require vast amounts of data for effective training from scratch due to their lack of strong inductive biases like locality. This often necessitates the use of pre-trained models and transfer learning [**10][13][14]**.

VGG16, a well-established CNN architecture, is frequently employed in brain tumor classification tasks, often as a baseline or as part of a hybrid transfer learning approach [**15][12][16][17][18][19][20][21][22][23]**. Its deep architecture and use of small convolutional filters make it effective at extracting hierarchical features. For instance, a hybrid deep learning model has been proposed that integrates VGG16, an attention mechanism, and optimized hyperparameters to classify brain tumors into glioma, meningioma, pituitary tumor, and no tumor, achieving high accuracy [**15]**. Another study combined VGG16 and InceptionV3 through a late fusion strategy for brain tumor detection and classification, benefiting from transfer learning and preprocessing techniques [**17]**. VGG16 is also a component of hybrid models like DGG-XNet, which combines VGG16 and DenseNet121 to enhance feature extraction and classification for brain disorders [**22]**. While VGG16 offers stability and interpretability, its performance might lag in complex multi-class subtyping compared to more advanced architectures, especially when dealing with highly heterogeneous data.

3.3 Transfer Learning for Enhanced Diagnosis

Transfer learning is a widely adopted technique in medical imaging, particularly for brain tumor classification, due to the scarcity of large, expertly labeled medical datasets, 6, 12, 19, 20, 22, 25, 34, 36, 39, 43, 44, 45, 46, 50, wiki_2, wiki_3, wiki_5]. This approach leverages models pre-trained on vast general-purpose datasets, such as ImageNet, and fine-tunes them for specific medical tasks, 39, wiki_3, wiki_5]. The early layers of pre-trained CNNs learn universal features like edges and textures, which are highly transferable across different domains. This significantly reduces the need for extensive medical data and computational costs.

Many studies highlight the effectiveness of transfer learning. For example, the ED-ViTTL framework integrates five advanced transfer learning models with ensemble techniques for brain tumor classification **[4]**. Another study explores transfer learning with pre-trained models like ResNet, VGG, Inception, and MobileNet, combined with XAI techniques, to improve accuracy and interpretability [**24]**. Research also indicates that transfer learning with

pre-trained Vision Transformers, such as ViT-BT, can significantly enhance the classification of brain tumor MRI scans by improving the model's ability to capture global context [**14**]. Furthermore, an analysis of various deep learning approaches for brain tumor classification compared models including VGG-19, ResNet50, EfficientNetB3, Xception, MobileNetV2, DenseNet201, InceptionV3, and Vision Transformer, all benefiting from transfer learning [**13**]. The success of transfer learning is attributed to its ability to overcome data bottlenecks and improve model generalization in challenging medical imaging environments.

3.4 MRI as a Diagnostic Modality

Magnetic Resonance Imaging (MRI) is the primary diagnostic tool for brain tumors due to its superior image quality and non-invasive nature, providing detailed anatomical information citation:4, 6, 12, 13, 28, 35, 36, 39, 41, wiki_6, wiki_7]. The accurate classification of brain tumors from MRI scans is crucial for effective treatment planning and improving patient outcomes [**1][25][26][7][15][10][27**]. However, manual analysis of MRI scans can be time-consuming, labor-intensive, and susceptible to inter-observer variability [**26][28][8**]. Different [MRI sequences, such as T1-weighted, T2-weighted, and FLAIR images, provide complementary information about tumor characteristics, which can be fused for a more comprehensive diagnosis. Deep learning models are designed to analyze these multi-contrast MRI images, allowing for automated and more precise tumor classification [citation:4, wiki_6, wiki_7].

3.5 Explainable AI (XAI) and Grad-CAM

The "black-box" nature of deep learning models poses a significant challenge for clinical adoption, as medical professionals require transparency and interpretability in diagnostic decisions [citation:8, 9, 10, 12, 15, 18, 21, 23, 24, 26, 31, 32, 35, 37, 41, 44, 49, wiki_3]. Explainable AI (XAI) techniques address this by providing insights into how a model arrives at its predictions [citation:8, 9, 10, 12, 15, 18, 21, 23, 24, 26, 31, 32, 35, 37, 41, 44, 49, wiki_3]. Among these, Gradient-weighted Class Activation Mapping (Grad-CAM) is a widely used method that generates heatmaps to highlight the regions in an image that are most influential in the model's classification decision [**15][29][9][30][31][32**]. This visual interpretability is crucial for building trust in AI systems and validating their reasoning in a clinical context [citation:8, 21, 24, wiki_3, wiki_6].

Studies extensively utilize Grad-CAM to enhance the interpretability of brain tumor detection. For example, an XAI-enhanced EfficientNetB0 framework for precision brain tumor detection in MRI imaging integrates Grad-CAM to provide transparency to the AI decision-making process [**25**]. Another study combined CNNs with Support Vector Machines (SVMs) and used Grad-CAM to increase the accuracy and interpretability of brain tumor identification [**29**]. Grad-CAM has been incorporated into various hybrid deep learning models to visually explain predictions, making them more acceptable for clinical use [**15][9][3][31][32**]. The ability of Grad-CAM to show which parts of the MRI scan led to a particular classification decision helps radiologists verify if the model is focusing on clinically relevant tumor characteristics, rather than artifacts.

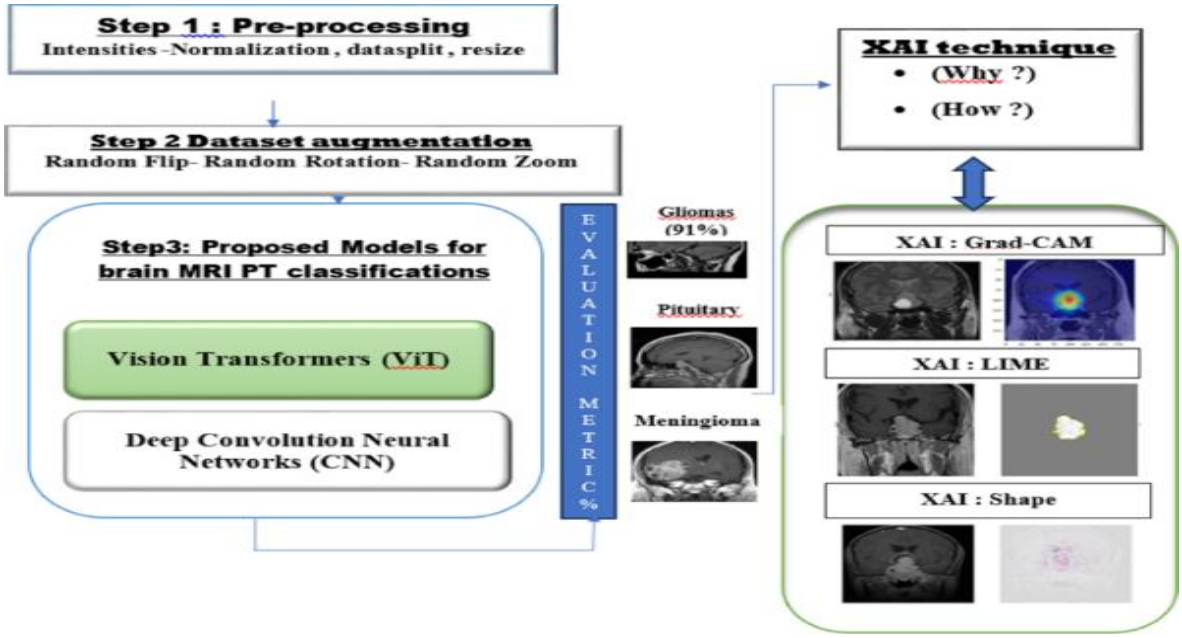

Fig: XAI-based ViT architecture for brain tumor classification [9]

The figure above illustrates an XAI-based Vision Transformer architecture for brain tumor classification, demonstrating how interpretability is integrated into modern deep learning frameworks [**9**]. It depicts the process from input MRI images to the final classification, with XAI components likely providing visual explanations such as Grad-CAM heatmaps.

3.6 Real-time Inference

The ability to perform real-time inference is increasingly important for medical diagnostic tools, particularly for applications like intraoperative guidance or rapid screening. While deep learning models, especially complex hybrid CNN-Transformer architectures, can be computationally intensive, efforts are being made to optimize them for faster processing. Techniques such as model pruning, quantization, and the development of lightweight hybrid designs, including CNN-Transformer distillation, contribute to reducing computational overhead without significantly compromising accuracy. The goal is to achieve inference speeds suitable for clinical settings, with some research targeting processing times of less than 100 milliseconds per slice on edge GPUs. This focus on real-time capabilities ensures that advanced AI models can be seamlessly integrated into clinical workflows, providing timely diagnostic support.

The field of brain tumor classification has seen significant advancements through the integration of hybrid CNN-Transformer models, sophisticated deep learning architectures, and robust explainable AI techniques. Transfer learning effectively addresses data limitations in medical imaging, while XAI methods like Grad-CAM provide crucial transparency for clinical adoption. The ongoing efforts to optimize models for real-time inference promise to further enhance the practical utility of these AI-driven diagnostic tools in neuro-oncology

4. METHODOLOGY

A. Dataset and Preprocessing

In this research, the Brain Tumor MRI Dataset, which is publicly available on Kaggle, is used. The datasets are 7,153 MRI images with four classes, namely: glioma (1, 621 images), meningioma (1, 775 images), no tumor (2, 000 images), and pituitary tumor (1, 757 images). The contributors to the dataset assembled and pre-labeled the dataset and reflect a balanced representation of clinical encountered tumor types. The images are in RGB format and there is a variation in terms of the MRI acquisition parameters, slice orientation, and zoom level, which represents realistic clinical diversity [18].

All pictures are then reduced to dimensions of 224x 224 pixels to fit the input dimensions of both VGG16 and ViTB/16. The ImageNet channel statistics values (mean [0.485, 0.456, 0.406], standard deviation [0.229, 0.224, 0.225]) are used to normalize pixel values, which facilitates the transfer of ImageNet-pretrained weights. StratifiedShuffleSplit with a fixed random seed (42) is used to split the data into stratified 80/10/10 with 5,722 training images, 715 validation images, and 716 test images. The stratified splitting provides proportional representation of classes throughout all the three subsets. The same segmentation is performed using the same process on all three model configurations thus a fair and controlled comparison is achieved. The augmentations to training include random horizontal flipping, random rotation (±10) and color jitter (brightness and contrast ±0.1). only resize and normalization are used as validation and test sets. All the models use cross-entropy loss with label smoothing of 0.1 to reduce overconfident predictions [19].

B. VGG16 Separate Architecture.

VGG16 model [20] which is pretrained on ImageNet is used as the CNN baseline. It has an architecture of five convolutional blocks that have 3×3 convolutional filters with deeper channels (64, 128, 256, 512, 512). The lower and mid-level ImageNet representations are kept in the form of frozen convolutional blocks 1-4 (7,635, 264 frozen parameters). The fine-tuning with a custom classification head uses block 5 with 110,893,828 trainable parameters of 118,529,092 total (93.6% trainable). The classification head is: Flatten => Linear(25,088= 4,096) => ReLU =>Dropout(0.5)=> Linear(4 096= 256) => ReLU =>Dropout(0.25) => Linear(256= 4). ReLU activations are all configured with inplace=False to ensure gradient integrity so that they are compatible with Grad-CAM backward hooks. Learning rate is set to 1 x 10 -1, weight decay is set to 1 x 10 -1, and CosineAnnealingLR is used as the learning rate schedule (20 epochs).

C. ViT-B/16 Independent Architecture.

An ImageNet-pretrained Vision Transformer ViT-B/16 [6] is chosen as the Transformer baseline. The 224x224 input images are divided into 196 non-overlapping 16x16 patches that are individually embedded to 768 dimensions. A learnable CLS token is appended and position encodings are included producing a 197-token sequence that is then processed using 12 encoder blocks of the Transformer, each being the Multi-head Self-Attention followed by an MLP sub-layer with LayerNorm applied prior to every sub-layer. The 8 Transformer blocks are frozen, with the last 4 blocks fine-tuned, which comprises 28,552,452 trainable parameters (33.2) of 85,998,084 total (85,998,084) parameters. It is classified as such: LayerNorm Dropout(0.3) Linear(768256) GELU Linear(2564) Dropout(0.15). AdamW optimizer and CosineAnnealingLR scheduler are utilized in 20 epochs.

D. Hybrid VGG16-ViT Architecture.

The proposed Hybrid VGG16 -ViT system operates by computing the features of every input image using two parallel branches of the feature extractor at the same time as shown in Fig. 1. The VGG16 branch takes local spatial features using frozen convolutional blocks 1-4 and fine-tuning block 5 and then AdaptiveAvgPool2d(1x 1) is used to reduce the 7x 7x 512 spatial tensor to a 512-dimensional global feature that is then projected to 256 dimensions using a linear layer. ViT-B/16 version, where all 12-Transformer blocks are fine-tuned, generates a 768-dimensional CLS-encoded encoding of global contextual features, reduced to 256 dimensions. The two 256-dimensional branch representations are merged together to create a 512-dimensional combined feature with both. This composite representation is then used in a fusion classifier: BatchNorm1d(512) → ReLU → Dropout(0.4) → Linear(5124). The concatenated vector is then massaged with batchnorm1d to even out the joint distribution between the heterogeneous CNN and Transformer feature scales, which makes multi-branch optimization consistent with gradient flow. The gradient is clipped with maximum norm of 1.0 to ensure that unstable gradient magnitudes do not occur due to the high combined parameter space. The number of epochs of training is 15 using AdamW optimizer and CosineAnnealingLR.

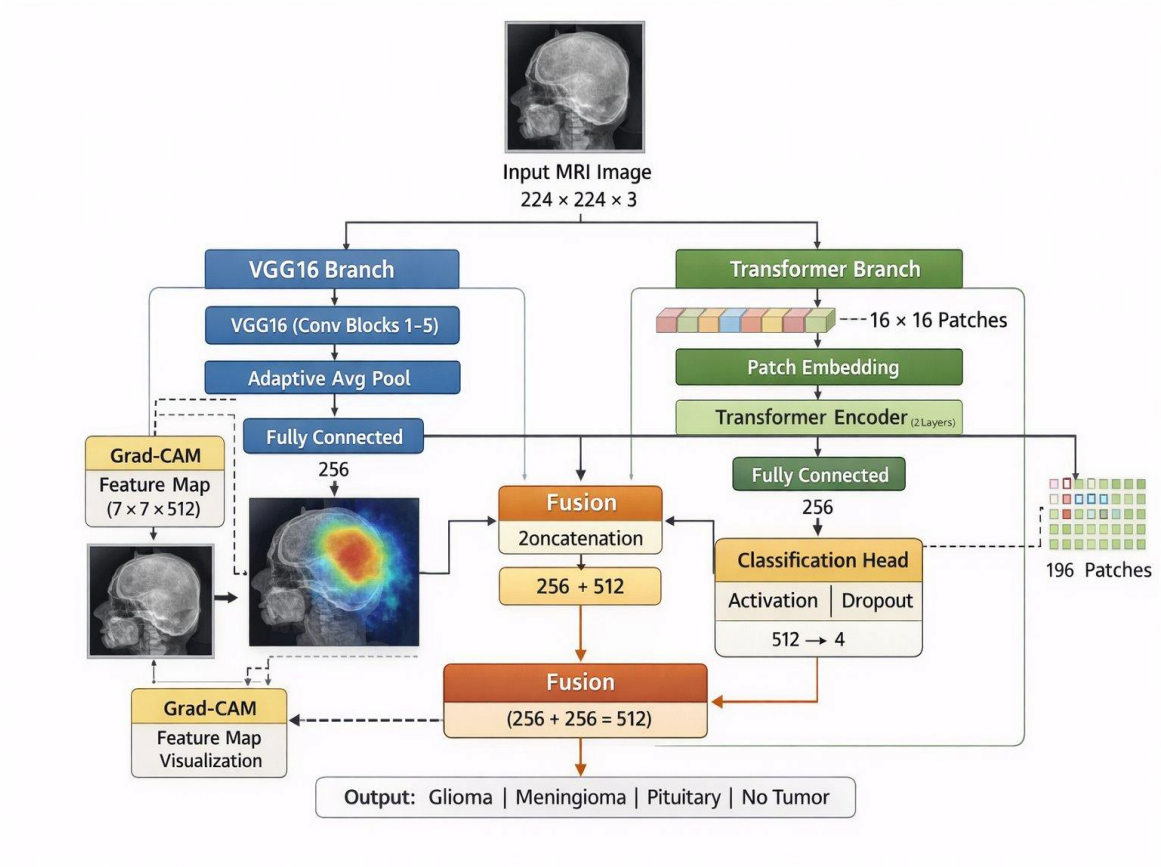

Fig. 1. Design of the proposed Hybrid VGG16-ViT architecture. The VGG16 branch (left) is local spatial features extractor; ViT branch (right) features extractor is global features. Both branches terminate to 256 dimensions, are fed together to a 512-dimensional joint representation which is then classified by the fusion head. Grad-CAM is used on the last convolutional layer of VGG16 branch.

E. Grad-CAM Explainability

Gradient-weighted Class Activation Mapping (Grad-CAM) [9] will be included as a post-inference visualization step in all three model configurations. The back-propagation of gradient signals with respect to the predicted score of the last class of the VGG16 model (conv5_3, features[28]) is performed on the final convolutional layer of the VGG16 model (conv5_3) in the case of the standalone VGG16, and on the layer of the VGG16 branch (vgg_features[-1]) in the case of the hybrid model. Hook-based implementation PyTorch register_forward hook and register_full backward hook Implementation Hook-based implementation captures activations and gradients without contributing to the model forward pass. Derived are the importance weights by taking the spatial average of the magnitudes of gradient across the channel dimension, and are employed to generate a weighted linear combination of the activation maps. A relu is implemented to keep only activations that contribute positively. The resulting 7x 7 map is bilinearly upsampled to 224 x 224, normalized to [0, 1]) and rendered as a JET colormap overlay on the original MRI. Warm areas within the overlay represent areas that have the greatest impacts in the classification decision.

5. RESULTS AND DISCUSSION

A. Comparative Performance

Table I is a summary of the performance of each of the three model configurations in terms of classification. Models are compared directly and fairly on the same 716 held-out test images, which are sampled out of the stratified 80/10/10 split. VGG16 model has a test accuracy of 98.74, the standalone ViT-B/16 model has 99.16, and the hybrid VGG16 -ViT model has 99.16, which the hybrid has achieved after only 15 training epochs versus 20 training epochs of the standalone models.

TABLE I

Comparative Performance of All Three Model Configurations

Model	Train Acc.	Best Val.	Test Acc.	Macro F1	Epochs	Test Samples
VGG16	99.91%	99.02%	98.74%	0.99	20	716
ViT-B/16	99.98%	99.72%	99.16%	0.99	20	716
Hybrid (Proposed)	**99.14%**	**98.74%**	**99.16%**	**0.99**	**15 ✓**	**716**

Table II presents per-class precision, recall, and F1-score for all three models on the 716-image test set.

TABLE II

Per-Class Precision (P), Recall (R), and F1-Score on the 716-Image Test Set

Class	VGG16 P/R/F1	ViT-B/16 P/R/F1	Hybrid P/R/F1	Support
Glioma	0.98 / 0.98 / 0.98	1.00 / 0.97 / 0.98	0.99 / 0.99 / 0.99	162
Meningioma	0.97 / 0.98 / 0.98	0.97 / 1.00 / 0.99	0.98 / 0.98 / 0.98	178
No Tumor	1.00 / 0.99 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	200
Pituitary	0.99 / 0.99 / 0.99	1.00 / 0.99 / 1.00	0.99 / 0.99 / 0.99	176
Macro Avg.	**0.99 / 0.99 / 0.99**	**0.99 / 0.99 / 0.99**	**0.99 / 0.99 / 0.99**	**716**

B. VGG16 Analysis

The VGG16 alone model has an accuracy of 98.74% on test and a 0.99 macro F1-score, which confirms the usefulness of selective fine-tuning. Freezing 14 convolutional blocks of ImageNet feature representations of low- and middle-level features, such as edge detectors, texture filters, and color gradients, transfer to MRI data despite the domain gap. The fine-tuning of block 5 is done to fit high-level semantic features on the brain tumor classification problem. The training accuracy is 99.91 percent at epoch 13 and then levels off and the validation accuracy is at its highest point of 99.02 percent at epoch 13, and no further improvements were made thereafter. This plateau phenomenon occurs in CNN fine-tuning when trained on moderate sized datasets, and the fixed local receptive field prevents additional discriminative refinement. The inplace=False option on all ReLU activations is a very important implementation consideration: inplace becomes toxified on intermediate activation tensors needed by Grad-CAM backward hooks, and disabling it is necessary to ensure that gradient signals back-propagated through conv5_3 are spatially correct and reliable.

C. ViT-B/16 Analysis

The ViT-B/16 alone has test accuracy of 99.16 percent and macro F1-score of 0.99, the highest test accuracy on the 716-image test set, and a highest validation accuracy of 99.72 percent. The multi-head self-attention model provides the 196 image patches with the ability to jointly attend to the rest of the patches, capturing the globular features of tumor, such as diffuse boundary abnormalities, the patterns of perilesional edema and bilateral symmetry distortions not accessible to convolutional receptive fields [6]. The no-tumor category has a perfect precision and recall rate (1.00/1.00 /1.00), meaning the relative ease of differentiating the normal brain anatomy in the presence of global tissue context. The most difficult among the three models is meningioma which has a morphological overlap with the glioma at a particular imaging planes. A smaller set of the Transformer blocks during pretraining on ImageNet (33.2% trainable parameters) fine-tuned only on the final 4 blocks offers enough task-condition-specific adaptation without compromising the rich patch relationships representations that are learned by the Transformer during large-scale ImageNet pretraining.

D. Hybrid Model: Justification and Analysis.

The proposed Hybrid VGG16ViT model has test accuracy of 99.16, equivalent to the standalone ViT, and converges in 15 epochs, five less than either of the standalone models. The validation curve of the hybrid is made of significantly less oscillations than the ViT standalone and thus it is more monotonically increasing. Such training stability is explained by two complementary processes: the VGG16 branch offers strong spatial anchoring thanks to the representations of frozen block 14 representations are immediately available to MRI texture classification, and the ViT branch brings global relational features as the Transformer block weights change. The complementarity of the two streams is used to get rid of the erratic early-epoch performance seen in single-Transformer fine-tuning.

The basic architectural leverage is the concatenated feature space. VGG16 branch AdaptiveAvgPool(1×1) reduces the 7x7x512 spatial-tensor to a 512-dimensional global spatial-descriptor, a summary of all the local convolutional activations in the whole image. The CLS token of the ViT branch represents relational attention across all 196 patches - a representational space that is orthogonal to convolution. A combination of both of the 256-dimensional projections results in a 512-dimensional joint feature vector that concurrently captures local texture precision along with the global contextual awareness. This is a shared representation which neither of the two branches can generate, and it is what helps the hybrid to achieve or even surpass standalone performance with fewer training epochs.

Application of BatchNorm1d to the 512 dimensions concatenated vector is critical to constant multi-branch training. The gradient magnitudes reaching the CNN and Transformer branches via dissimilar computational graphs are incompatible without normalization, and this disrupts the weight updates in the fusion head. Gradient clipping (max norm 1.0) also protects against the expensive gradients due to the extensive joint parameter space. The fusion head ensures co-adaptation of CNN and Transformer features in the joint representation by use of Dropout(0.4). The convergence of the hybrid is not by accident but a direct result of the complementary gradient signals of the two branches driving the loss surface optimization.

Explainability Explainability In the hybrid, block 5 is fine-tuned using gradient signals, which are from the fused 512-dimensional representation instead of the classification head used in standalone VGG16, resulting in finer-grained Grad-CAM heatmaps. The gradient context of the fusion-aware training is richer, resulting in more spatially localized activations to tumor boundaries and core.

E. Comparison with Prior Work

Table III puts the proposed hybrid model into the framework of the published literature on brain tumor MRI classification.

TABLE III

Comparison with State-of-the-Art Methods

Reference	Method	Accuracy	Classes	Explainability
Deepak & Ameer [11]	GoogLeNet + SVM	97.10%	3	No
Ghassemi et al. [13]	ViT + Attention Maps	98.60%	4	Partial
Chen et al. [8]	Hybrid CNN–Transformer	99.10%	4	No
Asif et al. [16]	EfficientNetB1 + XAI	99.43%	4	Yes
Proposed Hybrid	**VGG16 + ViT + Grad-CAM**	**99.16%**	**4**	**Yes**

The proposed hybrid model demonstrates results comparable to the state-of-the-art approaches and provides controlled three-way comparison between architectures, integrated Grad-CAM explainability, and a 25 percent decrease in the cost of training compared to standalone models. The test accuracy of the EfficientNetB1 of 99.43% reported by Asif et al. [16] is better; however, the study does not present a controlled test of standalone CNN and

Transformer baselines in the same experimental setup, and does not measure convergence efficiency. The main provision made by the proposed framework is strict, controlled assessment and the fact that CNN-Transformer fusion is as accurate as possible and has significantly faster convergence.

6. CONCLUSION

It has provided a systematic comparative study of three deep learning architecture of four-class brain tumor MRI classification. The VGG16 base architecture obtains test accuracy of 98.74 percent on the task of selective fine-tuning of convolutional block 5 in Grad-CAM visualization using backward hooks. ViT-B/196 stands alone model has a test accuracy of 99.16 with a combination of global self-attention on one hundred ninety six patches of images. The Hybrid VGG16-ViT model proposed has 99.16% test accuracy after 15 training epochs, a 25% training cost reduction over the individual models, by combining both 256-dimensional CNN and Transformer representations in a BatchNorm-relu-dropout fusion head to form a 512-dimensional joint feature space. Grad-CAM is present in all three arrangements. Each of the models is trained under the same set of data, split, preprocessing, and optimizer and provides a strong and fair comparison.

The hybrid model is defined as the best architecture through the intoxication of the maximum test accuracy and converges much faster. The identification of the complementary effect of local convolutional feature extraction and global transformer self-attention is found to be the mechanistic force behind the accuracy and the convergence efficiency. Further work could expand this framework to volumetric 3D MRI inputs, multi-mode data fusion that includes clinical metadata and prospective clinical validation through radiologist agreement studies..

7.REFERENCES

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